**Table C: Determination of time period, T as a function of distance of the CM of a ring from the string, l**

| No of obs | Distance from the string to the | | Distance from the string to The CM of a ring $l=(l' + l'')/2$ (cm) | Time of 20 Oscillations $T'$ (sec) | Time period $T = T'/20$ (sec) |
|---|---|---|---|---|---|
| | Inner side of the ring $l'$ (cm) | Outer side of the ring $l''$ (cm) | | | |
| 1 | 4cm | 6.5cm | 5.25cm | 3m 7sec 187s | 9.35 |
| 2 | 3cm | 5.5cm | 4.25cm | 3m 4sec 184s | 9.2 |
| 3 | 2cm | 4.5cm | 3.25cm | 2m 59sec 179s | 8.95 |
| 4 | 7cm | 9.5cm | 8.25cm | 3m 18sec 198s | 9.9sec |
| 5 | 8cm | 10.5cm | 9.25cm | 3m 30sec 210s | 10.5sec |

**Table D: Determination of rigidity modulus, η**

| Two positions of the ring Considered | Value of $l_1$ and $l_2$ (cm) | | Value of $T_1$ and $T_2$ (sec) | | Rigidity modulus, $\eta = \frac{16\pi LM}{r^4} \frac{(l_2^2 - l_1^2)}{(T_2^2 - T_1^2)}$ (dyne/cm$^2$) | Mean value of $\eta$ (dyne/cm$^2$) |
|---|---|---|---|---|---|---|
| | $l_1$ | $l_2$ | $T_1$ | $T_2$ | | |
| 1 & 2 | 5.25 | 4.25 | 9.35 | 9.2 | $6.2266 \times 10^{11}$ | $3.33026 \times 10^{11}$ |
| 2 & 3 | 4.25 | 3.25 | 9.2 | 8.95 | $3.0386 \times 10^{11}$ | |
| 3 & 4 | 3.25 | 8.25 | 8.95 | 9.9 | $5.9029 \times 10^{11}$ | |
| 4 & 5 | 8.25 | 9.25 | 9.9 | 10.5 | $2.6284 \times 10^{11}$ | |
| 1 & 5 | 5.25 | 9.25 | 9.35 | 10.5 | $4.67097 \times 10^{11}$ | |

0.2872 0.05cm r = 0.05 L = 21.1cm M = 282gm

**PROCEDURE:**

1. Mass, $M$ of one ring is determined.
2. The length, $L$ of the wire is measured from the point of suspension to the point at which the wire is attached to the cylinder. The diameter and hence the radius, $r$ of the wire is measured by the screw gauge.
3. The two rings are placed on the rod at equal distances from the string with the help of the scale calibrated on the body of the rod. The distance of the inner and outer side of a ring from the string, $l'$ and $l''$ are read from the scale. The distance of the **CM** of a ring from the string is calculated as $l = (l' + l'')/2$.
4. The rod with the two rings is made to oscillate slowly in a horizontal plane about the string which should remain steady in its vertical position. The time for 20 oscillations is recorded and the time period, $T$ is determined.
5. Steps (3) and (4) are repeated by charging the positions of the ring on the rod to determine $T$ as a function of $l$. At least 5 sets of such $l - T$ readings should be taken in total.
6. Rigidity modulus, $\eta$ is calculated using Eq.(6) by considering the 1st and 2nd sets of $l - T$ readings, 2nd and 3rd sets of $l - T$ readings, and so on.
7. Finally, mean value of $\eta$ and the probable experimental error is calculated.

**OBSERVATIONS:**

Mass of one ring, **M** = . . . . . gm.

**Table A: Determination of the length of the wire**

| No. of observation | Length of the wire (cm) | Mean length L (cm) |
|---|---|---|
| 1 | 81 | 81.1cm |

81.2 cm

**Table B: Determination of the radius "r" of the wire**

Least count (l.c.) of the screw gauge = . . . . . cm 0.01mm mm

| No of Obs. | Main scale reading A (cm) | Circular scale reading (B) | Value of circular scale reading C = B X l.c. (cm) | Total reading A + C (cm) | Mean diameter d (cm) | Mean radius r (cm) |
|---|---|---|---|---|---|---|
| 1 | 1 mm | 10 | 10×0.01 mm = 0.1 mm | 1.1 mm = 0.1 cm | 0.1 cm | 0.05 cm |

# VALUES OF CONSTANT SUPPLIED

1. **Hall Effect Experiment:**
   Thickness of Ge Semiconductor = 0.5 mm

2. **Rigidity Experiment:**
   Mass of the Ring: Black = 184 gm
   Silver = 282 gm

3. **Decay of Current:**
   Resistance = 33kΩ
   Capacitance = 2200μF

4. **e/m Experiment:**
   Distance Between The Plates = 23 mm
   Length of each Plate = 11 mm
   Distance from the edge of the Plate to the Screen = 85 mm

**CALCULATION:**

Rigidity modulus, $\eta = \frac{16\pi LM}{r^4} \frac{(l_2^2 - l_1^2)}{(T_2^2 - T_1^2)}$ dyne/cm$^2$

1. For positions 1 & 2 : $\eta_1$ =
2. For positions 2 & 3 : $\eta_2$ =
3. For positions 3 & 4 : $\eta_3$ =
4. For positions 4 & 5 : $\eta_4$ =

**Average,** $\eta = \frac{\eta 1 + \eta 2 + \eta 3 + \eta 4}{4} =$

**EXPERIMENTAL:**

**RESULT:**

**PRECAUTIONS:**

I. Since the value of **r** occurs to the fourth power and that of **T** to the second power, their values should be determined very precisely and carefully.

II. The suspension wire must pass perpendicular to the axis of the cylinder.

III. During torsional oscillation of the cylinder, care should be taken to avoid pendulum type of oscillation of the cylinder.

**Table C: Determination of time period, T as a function of distance of the CM of a ring from the string, l**

| No of obs | Distance from the string to the | | Distance from the string to | Time of 20 | Time period |
|---|---|---|---|---|---|
| | Inner side of the ring $l'$ (cm) | Outer side of the ring $l''$ (cm) | The CM of a ring $l=(l'+l'')/2$ (cm) | Oscillations $T'$ (sec) | $T = T'/20$ (sec) |
| 1 | 4cm | 6.5cm | 5.25cm | | |
| 2 | 5cm | 7.5cm | 6.25cm | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |

**Table D: Determination of rigidity modulus, η**

| Two positions of the ring Considered | Value of $l_1$ and $l_2$ (cm) | | Value of $T_1$ and $T_2$ (sec) | | Rigidity modulus, $\eta = \frac{16\pi LM}{r^4}\frac{(l_2^2-l_1^2)}{(T_2^2-T_1^2)}$ (dyne/cm$^2$) | Mean value of $\eta$ (dyne/cm$^2$) |
|---|---|---|---|---|---|---|
| | $l_1$ | $l_2$ | $T_1$ | $T_2$ | | |
| 1 & 2 | | | | | | |
| 2 & 3 | | | | | | |
| 3 & 4 | | | | | | |
| 4 & 5 | | | | | | |
| 1 & 5 | | | | | | |

**PROCEDURE:**

1. Mass, $M$ of one ring is determined.
2. The length, $L$ of the wire is measured from the point of suspension to the point at which the wire is attached to the cylinder. The diameter and hence the radius, $r$ of the wire is measured by the screw gauge.
3. The two rings are placed on the rod at equal distances from the string with the help of the scale calibrated on the body of the rod. The distance of the inner and outer side of a ring from the string, $l'$ and $l''$ are read from the scale. The distance of the **CM** of a ring from the string is calculated as $l = (l' + l'')/2$.
4. The rod with the two rings is made to oscillate slowly in a horizontal plane about the string which should remain steady in its vertical position. The time for 20 oscillations is recorded and the time period, $T$ is determined.
5. Steps (3) and (4) are repeated by charging the positions of the ring on the rod to determine $T$ as a function of $l$. At least 5 sets of such $l - T$ readings should be taken in total.
6. Rigidity modulus, $\eta$ is calculated using Eq.(6) by considering the 1st and 2nd sets of $l - T$ readings, 2nd and 3rd sets of $l - T$ readings, and so on.
7. Finally, mean value of $\eta$ and the probable experimental error is calculated.

**OBSERVATIONS:**

Mass of one ring, **M** = . . . . . gm.

**Table A: Determination of the length of the wire**

| No. of observation | Length of the wire (cm) | Mean length L (cm) |
|---|---|---|
| 1 | 81 | 81.1cm |
| 2 | 81.2 cm | |

**Table B: Determination of the radius "r" of the wire**

Least count (1.c.) of the screw gauge = . . . . . 0.001 cm 0.01mm

| No of Obs. | Main scale reading A (cm) | Circular scale reading (B) | Value of circular scale reading C = B X l.c. (cm) | Total reading A + C (cm) | Mean diameter d (cm) | Mean radius r (cm) |
|---|---|---|---|---|---|---|
| 1 | 1mm | 10 | 10×0.01mm =0.1mm | 1.1mm =0.001cm | 0.001cm | 0.0005cm |

The **MI** of the system about the string is (4)

$$I = I_{rod} + 2I_{ring}$$

Where $I_{rod}$ is the **MI** of the rod and $I_{ring}$ is that of one ring about the string. Using the parallel axis theorem, $I_{ring}$ can be written as,

$$I = Ml^2 + I^0_{ring} \quad (5)$$

Where $I^0_{ring}$ is the **MI** of the ring about an axis parallel to the string and passing through its centre of mass (**CM**), $l$ is the distance of the **CM** from the point of suspension and $M$ is the mass of one ring. Now let $I_1$ and $I_2$ be the **MI** of the system corresponding to two different positions of the rings where they are placed at distances $l_1$ and $l_2$ from the point of suspension. Then

$$I_1 = I_{rod} + 2(Ml_1^2 + I^0_{ring})$$

$$I_2 = I_{rod} + 2(Ml_2^2 + I^0_{ring})$$

Let $T_1$ and $T_2$ be the corresponding time period of oscillation. Then using Eq. (3), we have

$$T_1^2 = \frac{8\pi L I_1}{\eta r^4}$$

$$T_2^2 = \frac{8\pi L I_2}{\eta r^4} \quad (6)$$

Subtracting,

$$(T_2^2 - T_1^2) = \frac{8\pi L}{\eta r^4}(I_2 - I_1) = \frac{16\pi LM}{\eta r^4}(l_2^2 - l_1^2)$$

$$\Rightarrow \eta = \frac{16\pi LM}{r^4}\frac{(l_2^2 - l_1^2)}{(T_2^2 - T_1^2)} \text{ dyne/cm}^2 \quad (6)$$

Thus η can be calculated from Eq. (6).

# EXPERIMENT 2

## Rigidity modulus by dynamical method

**AIM**: To determine the rigidity modulus of a wire by dynamical method

**APPARATUS**: Dynamical method apparatus, stop watch, screw gauge, slide caliper, meter scale

**THEORY**: When a heavy body is suspended by a wire, with the weight of the wire being negligible compared to that of the body, and is made to execute torsional oscillation about the axis of suspension, the time period of oscillation is given by,

$$T = 2\pi\sqrt{\frac{I}{C}} \qquad (1)$$

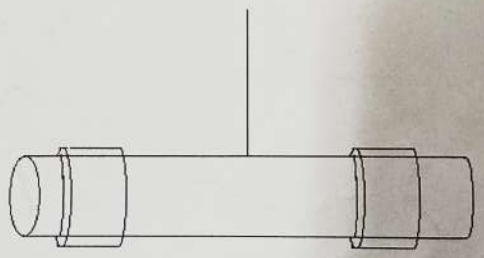
Where $I$ is the moment of inertia (**MI**) of the body about the axis of suspension and $C$ is the torque per unit twist.

If the cross-section of the wire be circular of area $\pi r^2$, $r$ being the radius of the wire, and $L$ the length of the wire, then

$$C = \frac{\pi\eta r^4}{2L} \qquad (2)$$

Where $\eta$ is the rigidity modulus of the material of the wire. From Eq. (1) and (2),

$$T^2 = \frac{8\pi L I}{\eta r^4} \qquad (3)$$

Therefore in principle, rigidity modulus $\eta$ can be calculated from Eq. (3) by determining other quantities in the equation. In the dynamical method, the need to determine $I$ is eliminated as accurate determination of **MI** of a body is often not possible.

The heavy body is taken in the form of a solid metallic cylindrical rod and is suspended from its center of mass by the string so that the rod lies horizontally. Two other identical heavy rings are attached to the solid rod in such a way that they are equidistant from the point of suspension, as shown in figure below.